Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Siegel, Peter H Ph.D.		
Project Title:	Nanoconverters: THz RF to DC Power Conversion for Driving Microdevices		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91109-8099	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:			
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	Nanostructures and nanosystems are flooding the research journals and are rapidly making their way into commercial systems. If we are to realize the full potential of these micromachines we must be able to supply power to individual devices without requiring wires or physical tethering. This task will provide direct direct current (DC) power for nanodevices and nanosystems using remote radio frequency (RF) generation, spatial transfer, and on-chip RF-to-DC conversion. We take advantage of state-of-the-art THz rectifier technology, micron sized directional antennas and monolithic GaAs MEMS (Micro-Electro-Mechanical Systems) style fabrication techniques to design and incorporate microminiature RF-to-DC converters directly onto nanostructures. Individually these microrectennas will supply as much as 1 mW of DC power in an area less than 50x50 mm2 (~ 40 W/cm2). Greater power and higher voltage can be obtained by simple series arrays (at the expense of physical space). Since the RF energy can penetrate without substantial loss through many plastics or even thin tissue, optical coupling is not required, i.e., energy can be transmitted into the skin with less loss than near InfraRed (IR) sources. The investigators plan to demonstrate the new technology by fabricating a		

Task Description:	working, remotely powered microchip. Applications are pervasive and include drive power for individual bioprobes, nanorobots, micromotors etc. as well as wireless power distribution to multiple chips or recharging of on-chip microbatteries. Many Biology Summer Research Program (BSRP) objectives will require remote or in-situ power for operating mechanical, biological or electronic devices, components or systems. The THz nanoconverter offers a useful and unique solution. The nanoconverter can bring DC power directly to a free roaming nanodevice that may be in a remote location or in a hostile environment. Although other possibilities exist (solar converters, micromotion converters, wires, fuel cells, etc.) none offer solutions that will work for all applications. The advantages of a THz-to-DC converter are numerous and efficiencies are expected to be high. We have already demonstrated RF-to-DC efficiencies of close to 20% with our 2.5 THz mixer circuits in a waveguide coupling environment. We expect to do at least as well with an optimized planar circuit. Advantages include small size, monolithic fabrication, close in directional coupling, minimal health risk, and better penetration through dielectrics than near IR or optical technologies. Payoffs span all branches of NASA technology including HEDS (health monitoring with nanoprobes), Bio-physical Research (intra-body probes and microrobots), Code S/Y (THz sensor development and optimization), and Code R (power beaming, nanolithography, nanostructures). This is a Technology Readiness Level (TRL) 1 task with enormous payoff and full prototype demonstration (TRL Level 3) at the end of two years. Remote power beaming for the purpose of providing energy to circuits without the need to propagate along wires or transmission lines is a concept as old as electric power generation itself. Since the early days of radio serious efforts to generate and distribute power via the airways has been an ongoing, although somewhat sporadic concern. Major breakthrough's in the field centered around developments of high power microwave generators like the klystron and TWT (traveling wave tube), high power handling capacity rectifiers, especially semiconductor diodes, and high efficiency power combining and rectifying receiver arrays or antennas. Recent performance at microwave frequencies (below 10 GHz) show system efficiencies above 80%. Although the market for microwave power transmission seems to consistently lag behind the development activities, there is still a serious effort going on within the NASA community to provide energy for such diverse applications as remote aircraft and balloon flights (as originally conceived by Brown), distributed actuator circuits for large reflector shaping, and remote power beaming from Earth to satellites or even from satellites to Earth. This task focuses on an application for remote power transmission that specifically addresses the problem of drive energy availability when ultra small devices are involved. We propose making use of THz RF signals and microminiature integrated antennas and rectifying diodes to efficiently mobilize nanodevices through power beaming. The technique eliminates locally situated self-contained power sources, communication lines and other physically tethered power transmission systems, and operates in a directive manner even in the presence of optically opaque substances. The use of THz signals, rather than the low frequency microwave radiation employed in previous work has significant advantages: (1) Small antenna dimensions due to the short wavelengths involved (consistent with integration of the receive & RF-to-DC converter elements directly onto microdevices), (2) High directivity with small focusing elements (allows directed energy beaming with hand-held lenses or mirrors), (3) Penetration through opaque materials such as plastic or skin (due to lower absorption in the THz range compared to optical or near infrared wavelengths in many materials), (4) Minimal damage to tissue or biological materials, since focused THz energy is more dispersed than shorter wavelengths (greater beam waist radius), (5) Direct RF-to-DC conversion in a fully integrated package (THz Schottky diodes integrated onto planar antennas) and, (6) Compatibility with monolithic processing (due to the small sizes involved and new processing capabilities of GaAs).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)